

Technology Stack & Software Requirements Specification

Date	15 March 2024
Team Member / Author	Naga Lakshmi
Project Name	Smart Lender: AI-Based Loan Eligibility Prediction System
Document Type	Technology Stack & Requirements Specification

1. Technology Stack (Tech Stack)

The Smart Lender system is built using a robust, decoupled modular architecture. It combines a heavy-duty Python-based machine learning pipeline with a lightweight web interface.

Layer / Component	Technology / Tool	Version	Description & Purpose
Programming Language	Python	3.10.x	Primary language for data preprocessing, model building, evaluation, and backend scripting.
Web Framework	Flask	3.0.x	Lightweight, flexible micro-framework used to host the web application, APIs, and dashboard routes.
Machine Learning	Scikit-Learn, XGBoost	Latest Stable	Used for training and evaluating models (Decision Tree, Random Forest, KNN, XGBoost) and scaling features.
Data Manipulation	Pandas, NumPy	Latest Stable	Enables high-performance computation, categorical encoding, cleaning, and matrix manipulations.
Frontend UI	HTML5, CSS3, JS, Bootstrap	v5.x	Builds the responsive web interface for applicants to submit details and views prediction dashboards.
Model Serialization	Joblib / Pickle	Standard Library	Used to serialize and save the best-performing model binary for production deployment runtime.

2. System Requirements (Software & Hardware)

2.1 Software Requirements

- **Operating System:** Windows 10/11, macOS Big Sur (or newer), or standard Linux distributions (Ubuntu/CentOS).
- **Python Runtime:** Python versions 3.8 to 3.11 installed with Pip package manager configured.
- **Web Browsers:** Modern browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, or Apple Safari with JavaScript enabled.
- **Integrated Dev Env (IDE):** VS Code, PyCharm, or Jupyter Notebooks (for preliminary model exploration and parameter tuning).

2.2 Hardware Requirements

Hardware Component	Minimum Requirement	Recommended Requirement
Processor (CPU)	Intel Core i3 / AMD Ryzen 3 (Dual-core or equivalent)	Intel Core i5 / AMD Ryzen 5 (Quad-core or better)
System Memory (RAM)	4 GB RAM	8 GB or 16 GB RAM (Highly recommended for executing heavy pipelines like XGBoost/RF)
Storage Space	2 GB Free Disk Space (for Python interpreter, dependencies, and datasets)	10 GB Free SSD Space (for local environments, serialized pipelines, historical logs, and datasets)

3. System Workflow Dependencies

1. Applicant Input / CSV Batch Upload (via Web Interface/HTML5 Form)
2. Data Processing Layer (Pandas cleaning & Scikit-Learn pipelines)
3. Inference Processing (Model loading via Joblib, predictions via XGBoost/Random Forest)
4. Output Delivery (JSON payload back to UI, interactive dashboards, and downloadable PDF reports).